1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC) tumors harboring a KRAS G12D mutation, constitutive KRAS signaling amplifies downstream MAPK/ERK and PI3K/AKT/mTOR pathways, driving a coordinated program of EMT/stemness and enhanced DNA damage response (DDR). This dual program promotes cisplatin resistance through (i) increased invasion and survival signaling and (ii) upregulated DNA repair gene activity (e.g., ERCC1, RAD51) that mitigates cisplatin-induced DNA crosslinks. Consequently, KRAS G12D-bearing MIBC cells exhibit higher basal MAPK/ERK activity, elevated EMT/stemness markers, and robust DDR gene expression, resulting in higher cisplatin IC50s compared with KRAS WT. Targeting KRAS effector pathways (MEK/ERK or PI3K/AKT/mTOR inhibitors) or directly inhibiting KRAS signaling will sensitize these cells to cisplatin, with combination regimens (cisplatin plus MEK or ERK inhibitors) showing synergistic anti-tumor effects in vitro and in vivo. This hypothesis predicts that KRAS G12D status will correlate with MAPK/EMT signatures and DDR readiness in patient samples and that isogenic models will recapitulate the resistance phenotype and its reversal upon pathway inhibition.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions